

Table 42: Simulation VIII: Empirical power and Monte Carlo standard error for Localized alternative: $x_0 = 0.25$, $\delta = 0.05$, $c = 0.5$, restricted to $r = 8, 10, 15$.

$r = 8$	Uniform	Central	Tail	Localized	$r = 10$	Uniform	Central	Tail	Localized	$r = 15$	Uniform	Central	Tail	Localized
50	0.0000 (0.0000)	0.0360 (0.0083)	0.0760 (0.0119)	0.0460 (0.0094)	50	0.0000 (0.0000)	0.0780 (0.0120)	0.0400 (0.0088)	0.0400 (0.0088)	50	0.0000 (0.0000)	0.0300 (0.0076)	0.0620 (0.0108)	0.0440 (0.0092)
100	0.0000 (0.0000)	0.0460 (0.0094)	0.0600 (0.0106)	0.0820 (0.0123)	100	0.0000 (0.0000)	0.0700 (0.0114)	0.0380 (0.0086)	0.0320 (0.0079)	100	0.0000 (0.0000)	0.0520 (0.0099)	0.0520 (0.0099)	0.0740 (0.0117)
200	0.0000 (0.0000)	0.0620 (0.0108)	0.0580 (0.0105)	0.0660 (0.0111)	200	0.0000 (0.0000)	0.0460 (0.0094)	0.0800 (0.0121)	0.0480 (0.0096)	200	0.0000 (0.0000)	0.0440 (0.0092)	0.0500 (0.0097)	0.0360 (0.0083)
500	0.0000 (0.0000)	0.0440 (0.0092)	0.0560 (0.0103)	0.0200 (0.0063)	500	0.0000 (0.0000)	0.0460 (0.0094)	0.0660 (0.0111)	0.0440 (0.0092)	500	0.0000 (0.0000)	0.0460 (0.0094)	0.0700 (0.0114)	0.0360 (0.0083)
1000	0.0000 (0.0000)	0.0340 (0.0081)	0.0420 (0.0090)	0.0400 (0.0088)	1000	0.0000 (0.0000)	0.0420 (0.0090)	0.0500 (0.0097)	0.0180 (0.0059)	1000	0.0000 (0.0000)	0.0660 (0.0111)	0.0560 (0.0103)	0.0440 (0.0092)

Table 43: Simulation VIII: Empirical power and Monte Carlo standard error for Localized alternative: $x_0 = 0.25$, $\delta = 0.05$, $c = 1.5$, restricted to $r = 8, 10, 15$.

$r = 8$	Uniform	Central	Tail	Localized	$r = 10$	Uniform	Central	Tail	Localized	$r = 15$	Uniform	Central	Tail	Localized
50	0.0000 (0.0000)	0.0240 (0.0068)	0.0560 (0.0103)	0.0460 (0.0094)	50	0.0000 (0.0000)	0.0820 (0.0123)	0.0340 (0.0081)	0.0620 (0.0108)	50	0.0000 (0.0000)	0.0600 (0.0106)	0.0400 (0.0088)	0.0380 (0.0086)
100	0.0000 (0.0000)	0.0640 (0.0109)	0.0640 (0.0109)	0.0440 (0.0092)	100	0.0000 (0.0000)	0.0700 (0.0114)	0.0400 (0.0088)	0.0340 (0.0081)	100	0.0000 (0.0000)	0.0360 (0.0083)	0.0340 (0.0081)	0.0640 (0.0109)
200	0.0000 (0.0000)	0.0600 (0.0106)	0.0600 (0.0106)	0.0620 (0.0108)	200	0.0000 (0.0000)	0.0520 (0.0099)	0.0440 (0.0092)	0.0660 (0.0111)	200	0.0000 (0.0000)	0.0500 (0.0097)	0.0380 (0.0086)	0.0480 (0.0096)
500	0.0000 (0.0000)	0.0360 (0.0083)	0.0240 (0.0068)	0.0320 (0.0079)	500	0.0000 (0.0000)	0.0340 (0.0081)	0.0560 (0.0103)	0.0320 (0.0079)	500	0.0000 (0.0000)	0.0500 (0.0097)	0.0480 (0.0096)	0.0240 (0.0068)
1000	0.0000 (0.0000)	0.0340 (0.0081)	0.0460 (0.0094)	0.0240 (0.0068)	1000	0.0000 (0.0000)	0.0520 (0.0099)	0.0620 (0.0108)	0.0260 (0.0071)	1000	0.0000 (0.0000)	0.0640 (0.0109)	0.0420 (0.0096)	0.0460 (0.0094)

Table 44: Simulation VIII: Empirical power and Monte Carlo standard error for Localized alternative: $x_0 = 0.25$, $\delta = 0.1$, $c = 0.5$, restricted to $r = 8, 10, 15$.

$r = 8$	Uniform	Central	Tail	Localized	$r = 10$	Uniform	Central	Tail	Localized	$r = 15$	Uniform	Central	Tail	Localized
50	0.0000 (0.0000)	0.0500 (0.0097)	0.0440 (0.0092)	0.0420 (0.0090)	50	0.0000 (0.0000)	0.0720 (0.0116)	0.0520 (0.0099)	0.0340 (0.0081)	50	0.0000 (0.0000)	0.0240 (0.0068)	0.0300 (0.0076)	0.0520 (0.0099)
100	0.0000 (0.0000)	0.0680 (0.0113)	0.0380 (0.0086)	0.0300 (0.0076)	100	0.0000 (0.0000)	0.0860 (0.0125)	0.0440 (0.0092)	0.0380 (0.0086)	100	0.0000 (0.0000)	0.0500 (0.0097)	0.0520 (0.0099)	0.0440 (0.0092)
200	0.0000 (0.0000)	0.0660 (0.0111)	0.0720 (0.0116)	0.0440 (0.0092)	200	0.0000 (0.0000)	0.0440 (0.0092)	0.0660 (0.0111)	0.0360 (0.0083)	200	0.0000 (0.0000)	0.0440 (0.0092)	0.0360 (0.0083)	0.0560 (0.0103)
500	0.0000 (0.0000)	0.0580 (0.0105)	0.0540 (0.0101)	0.0320 (0.0079)	500	0.0000 (0.0000)	0.0400 (0.0088)	0.0400 (0.0088)	0.0280 (0.0074)	500	0.0000 (0.0000)	0.0460 (0.0094)	0.0680 (0.0113)	0.0420 (0.0090)
1000	0.0000 (0.0000)	0.0500 (0.0097)	0.0560 (0.0103)	0.0580 (0.0105)	1000	0.0000 (0.0000)	0.0440 (0.0092)	0.0560 (0.0108)	0.0440 (0.0094)	1000	0.0000 (0.0000)	0.1000 (0.0134)	0.0540 (0.0101)	0.0420 (0.0090)

Table 45: Simulation VIII: Empirical power and Monte Carlo standard error for Localized alternative: $x_0 = 0.25$, $\delta = 0.1$, $c = 1.5$, restricted to $r = 8, 10, 15$.

$r = 8$	Uniform	Central	Tail	Localized	$r = 10$	Uniform	Central	Tail	Localized	$r = 15$	Uniform	Central	Tail	Localized
50	0.0000 (0.0000)	0.0440 (0.0092)	0.0500 (0.0097)	0.0340 (0.0081)	50	0.0000 (0.0000)	0.0820 (0.0123)	0.0260 (0.0071)	0.0300 (0.0076)	50	0.0000 (0.0000)	0.0580 (0.0105)	0.0540 (0.0101)	0.0620 (0.0108)
100	0.0000 (0.0000)	0.0600 (0.0106)	0.0420 (0.0090)	0.0180 (0.0059)	100	0.0000 (0.0000)	0.0580 (0.0105)	0.0300 (0.0076)	0.0380 (0.0086)	100	0.0000 (0.0000)	0.0440 (0.0092)	0.0400 (0.0088)	0.0560 (0.0103)
200	0.0000 (0.0000)	0.0480 (0.0096)	0.0420 (0.0090)	0.0440 (0.0092)	200	0.0000 (0.0000)	0.0500 (0.0097)	0.0460 (0.0094)	0.0440 (0.0092)	200	0.0000 (0.0000)	0.0540 (0.0101)	0.0440 (0.0092)	0.0680 (0.0113)
500	0.0000 (0.0000)	0.0480 (0.0096)	0.0500 (0.0097)	0.0400 (0.0088)	500	0.0000 (0.0000)	0.0500 (0.0097)	0.0600 (0.0106)	0.0220 (0.0066)	500	0.0000 (0.0000)	0.0480 (0.0096)	0.0640 (0.0109)	0.0380 (0.0086)
1000	0.0000 (0.0000)	0.0540 (0.0101)	0.0540 (0.0101)	0.0420 (0.0090)	1000	0.0000 (0.0000)	0.0340 (0.0081)	0.0620 (0.0108)	0.0360 (0.0083)	1000	0.0000 (0.0000)	0.0820 (0.0123)	0.0660 (0.0111)	0.0460 (0.0094)

Table 46: Simulation VIII: Empirical power and Monte Carlo standard error for Localized alternative: $x_0 = 0.25$, $\delta = 0.2$, $c = 0.5$, restricted to $r = 8, 10, 15$.

$r = 8$	Uniform	Central	Tail	Localized	$r = 10$	Uniform	Central	Tail	Localized	$r = 15$	Uniform	Central	Tail	Localized
50	0.0000 (0.0000)	0.0540 (0.0101)	0.0640 (0.0109)	0.0380 (0.0086)	50	0.0000 (0.0000)	0.0680 (0.0113)	0.0340 (0.0081)	0.0500 (0.0097)	50	0.0000 (0.0000)	0.0500 (0.0097)	0.0360 (0.0083)	0.0440 (0.0092)
100	0.0000 (0.0000)	0.0700 (0.0114)	0.0500 (0.0097)	0.0460 (0.0094)	100	0.0000 (0.0000)	0.0700 (0.0114)	0.0520 (0.0099)	0.0460 (0.0094)	100	0.0000 (0.0000)	0.0360 (0.0083)	0.0480 (0.0096)	0.0580 (0.0105)
200	0.0000 (0.0000)	0.0320 (0.0079)	0.0520 (0.0099)	0.0300 (0.0076)	200	0.0000 (0.0000)	0.0620 (0.0108)	0.0860 (0.0125)	0.0680 (0.0113)	200	0.0000 (0.0000)	0.0580 (0.0105)	0.0440 (0.0092)	0.0380 (0.0086)
500	0.0000 (0.0000)	0.0440 (0.0092)	0.0560 (0.0103)	0.0520 (0.0099)	500	0.0000 (0.0000)	0.0480 (0.0096)	0.0580 (0.0105)	0.0280 (0.0074)	500	0.0000 (0.0000)	0.0500 (0.0097)	0.0800 (0.0121)	0.0280 (0.0071)
1000	0.0000 (0.0000)	0.0520 (0.0099)	0.0420 (0.0090)	0.0480 (0.0096)	1000	0.0000 (0.0000)	0.0440 (0.0092)	0.0580 (0.0105)	0.0480 (0.0096)	1000	0.0000 (0.0000)	0.0760 (0.0119)	0.0400 (0.0088)	0.0440 (0.0092)

Table 47: Simulation VIII: Empirical power and Monte Carlo standard error for Localized alternative: $x_0 = 0.25$, $\delta = 0.2$, $c = 1.5$, restricted to $r = 8, 10, 15$.

$r = 8$	Uniform	Central	Tail	Localized	$r = 10$	Uniform	Central	Tail	Localized	$r = 15$	Uniform	Central	Tail	Localized
50	0.0000 (0.0000)	0.0460 (0.0094)	0.0640 (0.0109)	0.0280 (0.0074)	50	0.0000 (0.0000)	0.0660 (0.0111)	0.0500 (0.0097)	0.0660 (0.0111)	50	0.0000 (0.0000)	0.0420 (0.0090)	0.0420 (0.0090)	0.0340 (0.0081)
100	0.0000 (0.0000)	0.0620 (0.0108)	0.0440 (0.0092)	0.0540 (0.0101)	100	0.0000 (0.0000)	0.0640 (0.0109)	0.0460 (0.0094)	0.0580 (0.0105)	100	0.0000 (0.0000)	0.0340 (0.0081)	0.0380 (0.0086)	0.0640 (0.0109)
200	0.0000 (0.0000)	0.0600 (0.0106)	0.0560 (0.0103)	0.0300 (0.0076)	200	0.0000 (0.0000)	0.0540 (0.0101)	0.0620 (0.0108)	0.0640 (0.0109)	200	0.0000 (0.0000)	0.0320 (0.0079)	0.0400 (0.0088)	0.0360 (0.0083)
500	0.0000 (0.0000)	0.0440 (0.0092)	0.0540 (0.0101)	0.0320 (0.0079)	500	0.0000 (0.0000)	0.0360 (0.0083)	0.0780 (0.0120)	0.0240 (0.0068)	500	0.0000 (0.0000)	0.0500 (0.0097)	0.0720 (0.0116)	0.0160 (0.0056)
1000	0.0000 (0.0000)	0.0380 (0.0086)	0.0540 (0.0101)	0.0340 (0.0081)	1000	0.0000 (0.0000)	0.0480 (0.0096)	0.0640 (0.0109)	0.0460 (0.0094)	1000	0.0000 (0.0000)	0.0780 (0.0120)	0.0580 (0.0105)	0.0400 (0.0088)

Table 48: Simulation VIII: Empirical power and Monte Carlo standard error for Localized alternative: $x_0 = 0.5$, $\delta = 0.05$, $c = 0.5$, restricted to $r = 8, 10, 15$.

$r = 8$	Uniform	Central	Tail	Localized	$r = 10$	Uniform	Central	Tail	Localized	$r = 15$	Uniform	Central	Tail	Localized
50	0.0000 (0.0000)	0.0380 (0.0086)	0.0620 (0.0108)	0.0240 (0.0068)	50	0.0000 (0.0000)	0.0900 (0.0128)	0.0400 (0.0088)	0.0280 (0.0074)	50	0.0000 (0.0000)	0.0600 (0.0106)	0.0340 (0.0081)	0.0500 (0.0097)
100	0.0000 (0.0000)	0.0640 (0.0109)	0.0360 (0.0083)	0.0180 (0.0059)	100	0.0000 (0.0000)	0.0640 (0.0109)	0.0320 (0.0079)	0.0560 (0.0103)	100	0.0000 (0.0000)	0.0460 (0.0094)	0.0600 (0.0106)	0.0400 (0.0088)
200	0.0000 (0.0000)	0.0460 (0.0094)	0.0720 (0.0116)	0.0380 (0.0086)	200	0.0000 (0.0000)	0.0440 (0.0092)	0.0520 (0.0099)	0.0460 (0.0094)	200	0.0000 (0.0000)	0.0480 (0.0096)	0.0440 (0.0092)	0.0300 (0.0076)
500	0.0000 (0.0000)	0.0440 (0.0092)	0.0440 (0.0092)	0.0660 (0.0111)	500	0.0000 (0.0000)	0.0320 (0.0079)	0.0540 (0.0101)	0.0640 (0.0109)	500	0.0000 (0.0000)	0.0380 (0.0086)	0.0600 (0.0106)	0.0560 (0.0103)
1000	0.0000 (0.0000)	0.0420 (0.0090)	0.0540 (0.0101)	0.0520 (0.0099)	1000	0.0000 (0.0000)	0.0500 (0.0097)	0.0540 (0.0101)	0.0460 (0.0094)	1000	0.0000 (0.0000)	0.0820 (0.0123)	0.0540 (0.0101)	0.0400 (0.0088)

Table 49: Simulation VIII: Empirical power and Monte Carlo standard error for Localized alternative: $x_0 = 0.5$, $\delta = 0.05$, $c = 1.5$, restricted to $r = 8, 10, 15$.

$r = 8$	Uniform	Central	Tail	Localized	$r = 10$	Uniform	Central	Tail	Localized	$r = 15$	Uniform	Central	Tail	Localized
50	0.0000 (0.0000)	0.0380 (0.0086)	0.0580 (0.0105)	0.0360 (0.0083)	50	0.0000 (0.0000)	0.0720 (0.0116)	0.0440 (0.0092)	0.0660 (0.0111)	50	0.0000 (0.0000)	0.0540 (0.0101)	0.0460 (0.0094)	0.0540 (0.0101)
100	0.0000 (0.0000)	0.0400 (0.0088)	0.0660 (0.0111)	0.0220 (0.0066)	100	0.0000 (0.0000)	0.0760 (0.0119)	0.0320 (0.0079)	0.0300 (0.0076)	100	0.0000 (0.0000)	0.0340 (0.0081)	0.0360 (0.0083)	0.0440 (0.0092)
200	0.0000 (0.0000)	0.0420 (0.0090)	0.0420 (0.0090)	0.0420 (0.0090)	200	0.0000 (0.0000)	0.0320 (0.0079)	0.0640 (0.0109)	0.0380 (0.0086)	200	0.0000 (0.0000)	0.0300 (0.0076)	0.0320 (0.0079)	0.0420 (0.0090)
500	0.0000 (0.0000)	0.0540 (0.0101)	0.0540 (0.0101)	0.0440 (0.0092)	500	0.0000 (0.0000)	0.0500 (0.0097)	0.0880 (0.0127)	0.0500 (0.0097)	500	0.0000 (0.0000)	0.0400 (0.0088)	0.0740 (0.0117)	0.0420 (0.0090)
1000	0.0000 (0.0000)	0.0480 (0.0096)	0.0580 (0.0105)	0.0660 (0.0111)	1000	0.0000 (0.0000)	0.0440 (0.0092)	0.0700 (0.0114)	0.0500 (0.0097)	1000	0.0000 (0.0000)	0.0860 (0.0125)	0.0480 (0.0096)	0.0360 (0.0083)

Table 50: Simulation VIII: Empirical power and Monte Carlo standard error for Localized alternative: $x_0 = 0.5$, $\delta = 0.1$, $c = 0.5$, restricted to $r = 8, 10, 15$.

$r = 8$	Uniform	Central	Tail	Localized	$r = 10$	Uniform	Central	Tail	Localized	$r = 15$	Uniform	Central	Tail	Localized
50	0.0000 (0.0000)	0.0400 (0.0088)	0.0460 (0.0094)	0.0420 (0.0090)	50	0.0000 (0.0000)	0.0660 (0.0111)	0.0460 (0.0094)	0.0300 (0.0076)	50	0.0000 (0.0000)	0.0440 (0.0092)	0.0400 (0.0088)	0.0380 (0.0086)
100	0.0000 (0.0000)	0.0500 (0.0097)	0.0440 (0.0092)	0.0460 (0.0094)	100	0.0000 (0.0000)	0.0580 (0.0105)	0.0460 (0.0094)	0.0580 (0.0105)	100	0.0000 (0.0000)	0.0440 (0.0092)	0.0500 (0.0097)	0.0820 (0.0123)
200	0.0000 (0.0000)	0.0560 (0.0103)	0.0320 (0.0079)	0.0560 (0.0103)	200	0.0000 (0.0000)	0.0420 (0.0090)	0.0580 (0.0105)	0.0420 (0.0090)	200	0.0000 (0.0000)	0.0220 (0.0066)	0.0420 (0.0090)	0.0460 (0.0094)
500	0.0000 (0.0000)	0.0400 (0.0088)	0.0340 (0.0081)	0.0360 (0.0083)	500	0.0000 (0.0000)	0.0400 (0.0088)	0.0660 (0.0111)	0.0540 (0.0101)	500	0.0000 (0.0000)	0.0440 (0.0092)	0.0580 (0.0105)	0.0600 (0.0106)
1000	0.0000 (0.0000)	0.0260 (0.0071)	0.0720 (0.0116)	0.0540 (0.0101)	1000	0.0000 (0.0000)	0.0380 (0.0086)	0.0580 (0.0105)	0.0340 (0.0081)	1000	0.0000 (0.0000)	0.0760 (0.0119)	0.0380 (0.0086)	0.0180 (0.0059)

Table 51: Simulation VIII: Empirical power and Monte Carlo standard error for Localized alternative: $x_0 = 0.5$, $\delta = 0.1$, $c = 1.5$, restricted to $r = 8, 10, 15$.

$r = 8$	Uniform	Central	Tail	Localized	$r = 10$	Uniform	Central	Tail	Localized	$r = 15$	Uniform	Central	Tail	Localized
50	0.0000 (0.0000)	0.0380 (0.0086)	0.0380 (0.0086)	0.0440 (0.0092)	50	0.0000 (0.0000)	0.0640 (0.0109)	0.0340 (0.0081)	0.0420 (0.0090)	50	0.0000 (0.0000)	0.0480 (0.0096)	0.0260 (0.0071)	0.0480 (0.0096)
100	0.0000 (0.0000)	0.0540 (0.0101)	0.0700 (0.0114)	0.0240 (0.0068)	100	0.0000 (0.0000)	0.0640 (0.0109)	0.0440 (0.0092)	0.0340 (0.0081)	100	0.0000 (0.0000)	0.0300 (0.0076)	0.0380 (0.0086)	0.0580 (0.0105)
200	0.0000 (0.0000)	0.0480 (0.0096)	0.0780 (0.0120)	0.0340 (0.0081)	200	0.0000 (0.0000)	0.0360 (0.0083)	0.0560 (0.0103)	0.0400 (0.0088)	200	0.0000 (0.0000)	0.0420 (0.0090)	0.0300 (0.0076)	0.0260 (0.0071)
500	0.0000 (0.0000)	0.0460 (0.0094)	0.0360 (0.0083)	0.0340 (0.0081)	500	0.0000 (0.0000)	0.0240 (0.0068)	0.0680 (0.0113)	0.0300 (0.0076)	500	0.0000 (0.0000)	0.0420 (0.0090)	0.0640 (0.0109)	0.0500 (0.0097)
1000	0.0000 (0.0000)	0.0320 (0.0079)	0.0660 (0.0111)	0.0500 (0.0097)	1000	0.0000 (0.0000)	0.0460 (0.0094)	0.0720 (0.0116)	0.0400 (0.0088)	1000	0.0000 (0.0000)	0.0680 (0.0113)	0.0460 (0.0094)	0.0260 (0.0071)

Table 52: Simulation VIII: Empirical power and Monte Carlo standard error for Localized alternative: $x_0 = 0.5$, $\delta = 0.2$, $c = 0.5$, restricted to $r = 8, 10, 15$.

$r = 8$	Uniform	Central	Tail	Localized	$r = 10$	Uniform	Central	Tail	Localized	$r = 15$	Uniform	Central	Tail	Localized
50	0.0000 (0.0000)	0.0460 (0.0094)	0.0500 (0.0097)	0.0300 (0.0076)	50	0.0000 (0.0000)	0.0700 (0.0114)	0.0320 (0.0079)	0.0760 (0.0119)	50	0.0000 (0.0000)	0.0600 (0.0106)	0.0220 (0.0066)	0.0620 (0.0108)
100	0.0000 (0.0000)	0.0340 (0.0081)	0.0560 (0.0103)	0.0700 (0.0114)	100	0.0000 (0.0000)	0.0680 (0.0113)	0.0420 (0.0090)	0.0440 (0.0092)	100	0.0000 (0.0000)	0.0440 (0.0092)	0.0360 (0.0083)	0.0620 (0.0108)
200	0.0000 (0.0000)	0.0520 (0.0099)	0.0560 (0.0103)	0.0300 (0.0076)	200	0.0000 (0.0000)	0.0480 (0.0096)	0.0620 (0.0108)	0.0360 (0.0083)	200	0.0000 (0.0000)	0.0500 (0.0097)	0.0420 (0.0090)	0.0200 (0.0063)
500	0.0000 (0.0000)	0.0500 (0.0097)	0.0440 (0.0092)	0.0480 (0.0096)	500	0.0000 (0.0000)	0.0300 (0.0076)	0.0500 (0.0097)	0.0600 (0.0106)	500	0.0000 (0.0000)	0.0400 (0.0088)	0.0420 (0.0090)	0.0440 (0.0092)
1000	0.0000 (0.0000)	0.0540 (0.0101)	0.0740 (0.0117)	0.0160 (0.0056)	1000	0.0000 (0.0000)	0.0220 (0.0066)	0.0780 (0.0120)	0.0620 (0.0108)	1000	0.0000 (0.0000)	0.0740 (0.0117)	0.0740 (0.0117)	0.0480 (0.0096)

Table 53: Simulation VIII: Empirical power and Monte Carlo standard error for Localized alternative: $x_0 = 0.5$, $\delta = 0.2$, $c = 1.5$, restricted to $r = 8, 10, 15$.

$r = 8$	Uniform	Central	Tail	Localized	$r = 10$	Uniform	Central	Tail	Localized	$r = 15$	Uniform	Central	Tail	Localized
50	0.0000 (0.0000)	0.0400 (0.0088)	0.0460 (0.0094)	0.0260 (0.0071)	50	0.0000 (0.0000)	0.0860 (0.0125)	0.0140 (0.0053)	0.0620 (0.0108)	50	0.0000 (0.0000)	0.0680 (0.0113)	0.0260 (0.0071)	0.0720 (0.0116)
100	0.0000 (0.0000)	0.0340 (0.0081)	0.0440 (0.0092)	0.0560 (0.0103)	100	0.0000 (0.0000)	0.0580 (0.0105)	0.0440 (0.0092)	0.0540 (0.0101)	100	0.0000 (0.0000)	0.0180 (0.0059)	0.0520 (0.0099)	0.0600 (0.0106)
200	0.0000 (0.0000)	0.0320 (0.0079)	0.0620 (0.0108)	0.0280 (0.0074)	200	0.0000 (0.0000)	0.0280 (0.0074)	0.0680 (0.0113)	0.0460 (0.0094)	200	0.0000 (0.0000)	0.0420 (0.0090)	0.0360 (0.0083)	0.0280 (0.0074)
500	0.0000 (0.0000)	0.0500 (0.0097)	0.0600 (0.0106)	0.0560 (0.0103)	500	0.0000 (0.0000)	0.0260 (0.0071)	0.0940 (0.0131)	0.0520 (0.0099)	500	0.0000 (0.0000)	0.0480 (0.0096)	0.0640 (0.0109)	0.0400 (0.0088)
1000	0.0000 (0.0000)	0.0240 (0.0068)	0.0700 (0.0114)	0.0240 (0.0068)	1000	0.0000 (0.0000)	0.0380 (0.0086)	0.0680 (0.0113)	0.0500 (0.0097)	1000	0.0000 (0.0000)	0.0540 (0.0101)	0.0540 (0.0101)	0.0440 (0.0092)

Table 54: Simulation VIII: Empirical power and Monte Carlo standard error for Localized alternative: $x_0 = 0.75$, $\delta = 0.05$, $c = 0.5$, restricted to $r = 8, 10, 15$.

$r = 8$	Uniform	Central	Tail	Localized	$r = 10$	Uniform	Central	Tail	Localized	$r = 15$	Uniform	Central	Tail	Localized
50	0.0000 (0.0000)	0.0460 (0.0094)	0.0680 (0.0113)	0.0760 (0.0119)	50	0.0000 (0.0000)	0.0740 (0.0117)	0.0360 (0.0083)	0.0600 (0.0106)	50	0.0000 (0.0000)	0.0380 (0.0086)	0.0540 (0.0101)	0.0400 (0.0088)
100	0.0000 (0.0000)	0.0500 (0.0097)	0.0520 (0.0099)	0.0520 (0.0099)	100	0.0000 (0.0000)	0.0540 (0.0101)	0.0560 (0.0103)	0.0440 (0.0092)	100	0.0000 (0.0000)	0.0440 (0.0092)	0.0400 (0.0088)	0.0340 (0.0096)
200	0.0000 (0.0000)	0.0540 (0.0101)	0.0680 (0.0113)	0.0400 (0.0088)	200	0.0000 (0.0000)	0.0420 (0.0090)	0.0760 (0.0119)	0.0460 (0.0094)	200	0.0000 (0.0000)	0.0300 (0.0076)	0.0540 (0.0101)	0.0580 (0.0105)
500	0.0000 (0.0000)	0.0700 (0.0114)	0.0400 (0.0088)	0.0500 (0.0097)	500	0.0000 (0.0000)	0.0460 (0.0094)	0.0640 (0.0109)	0.0540 (0.0101)	500	0.0000 (0.0000)	0.0420 (0.0090)	0.0500 (0.0097)	0.0560 (0.0103)
1000	0.0000 (0.0000)	0.0400 (0.0088)	0.0400 (0.0088)	0.0440 (0.0092)	1000	0.0000 (0.0000)	0.0220 (0.0060)	0.0660 (0.0111)	0.0560 (0.0103)	1000	0.0000 (0.0000)	0.0740 (0.0117)	0.0480 (0.0096)	0.0560 (0.0103)

Table 55: Simulation VIII: Empirical power and Monte Carlo standard error for Localized alternative: $x_0 = 0.75$, $\delta = 0.05$, $c = 1.5$, restricted to $r = 8, 10, 15$.

$r = 8$	Uniform	Central	Tail	Localized	$r = 10$	Uniform	Central	Tail	Localized	$r = 15$	Uniform	Central	Tail	Localized
50	0.0000 (0.0000)	0.0580 (0.0105)	0.0560 (0.0103)	0.0680 (0.0113)	50	0.0000 (0.0000)	0.0880 (0.0127)	0.0560 (0.0103)	0.0260 (0.0071)	50	0.0000 (0.0000)	0.0440 (0.0092)	0.0440 (0.0092)	0.0400 (0.0088)
100	0.0000 (0.0000)	0.0540 (0.0101)	0.0400 (0.0088)	0.0380 (0.0086)	100	0.0000 (0.0000)	0.0700 (0.0114)	0.0460 (0.0094)	0.0580 (0.0105)	100	0.0000 (0.0000)	0.0300 (0.0076)	0.0440 (0.0092)	0.0480 (0.0096)
200	0.0000 (0.0000)	0.0580 (0.0105)	0.0480 (0.0096)	0.0200 (0.0063)	200	0.0000 (0.0000)	0.0500 (0.0097)	0.0500 (0.0097)	0.0200 (0.0063)	200	0.0000 (0.0000)	0.0420 (0.0090)	0.0280 (0.0074)	0.0560 (0.0103)
500	0.0000 (0.0000)	0.0420 (0.0090)	0.0340 (0.0081)	0.0300 (0.0076)	500	0.0000 (0.0000)	0.0440 (0.0092)	0.0640 (0.0109)	0.0420 (0.0090)	500	0.0000 (0.0000)	0.0480 (0.0096)	0.0700 (0.0114)	0.0320 (0.0079)
1000	0.0000 (0.0000)	0.0480 (0.0096)	0.0740 (0.0117)	0.0200 (0.0063)	1000	0.0000 (0.0000)	0.0420 (0.0090)	0.0500 (0.0097)	0.0340 (0.0081)	1000	0.0000 (0.0000)	0.0740 (0.0117)	0.0600 (0.0106)	0.0680 (0.0113)

Table 56: Simulation VIII: Empirical power and Monte Carlo standard error for Localized alternative: $x_0 = 0.75$, $\delta = 0.1$, $c = 0.5$, restricted to $r = 8, 10, 15$.

$r = 8$	Uniform	Central	Tail	Localized	$r = 10$	Uniform	Central	Tail	Localized	$r = 15$	Uniform	Central	Tail	Localized
50	0.0000 (0.0000)	0.0560 (0.0103)	0.0740 (0.0117)	0.0480 (0.0096)	50	0.0000 (0.0000)	0.0860 (0.0125)	0.0360 (0.0083)	0.0600 (0.0106)	50	0.0000 (0.0000)	0.0400 (0.0088)	0.0420 (0.0090)	0.0160 (0.0056)
100	0.0000 (0.0000)	0.0620 (0.0108)	0.0600 (0.0106)	0.0860 (0.0125)	100	0.0000 (0.0000)	0.0720 (0.0116)	0.0480 (0.0096)	0.0360 (0.0083)	100	0.0000 (0.0000)	0.0340 (0.0081)	0.0460 (0.0094)	0.0260 (0.0071)
200	0.0000 (0.0000)	0.0680 (0.0113)	0.0500 (0.0097)	0.0620 (0.0108)	200	0.0000 (0.0000)	0.0500 (0.0097)	0.0620 (0.0108)	0.0500 (0.0097)	200	0.0000 (0.0000)	0.0320 (0.0079)	0.0420 (0.0090)	0.0480 (0.0096)
500	0.0000 (0.0000)	0.0380 (0.0086)	0.0560 (0.0103)	0.0460 (0.0094)	500	0.0000 (0.0000)	0.0500 (0.0097)	0.0600 (0.0106)	0.0560 (0.0103)	500	0.0000 (0.0000)	0.0800 (0.0121)	0.0520 (0.0099)	0.0320 (0.0079)
1000	0.0000 (0.0000)	0.0340 (0.0081)	0.0760 (0.0119)	0.0240 (0.0068)	1000	0.0000 (0.0000)	0.0440 (0.0092)	0.0680 (0.0113)	0.0560 (0.0103)	1000	0.0000 (0.0000)	0.0500 (0.0097)	0.0500 (0.0097)	0.0460 (0.0094)

Table 57: Simulation VIII: Empirical power and Monte Carlo standard error for Localized alternative: $x_0 = 0.75$, $\delta = 0.1$, $c = 1.5$, restricted to $r = 8, 10, 15$.

$r = 8$	Uniform	Central	Tail	Localized	$r = 10$	Uniform	Central	Tail	Localized	$r = 15$	Uniform	Central	Tail	Localized
50	0.0000 (0.0000)	0.0420 (0.0090)	0.0520 (0.0099)	0.0340 (0.0081)	50	0.0000 (0.0000)	0.0820 (0.0123)	0.0300 (0.0076)	0.0360 (0.0083)	50	0.0000 (0.0000)	0.0500 (0.0097)	0.0300 (0.0076)	0.0140 (0.0053)
100	0.0000 (0.0000)	0.0680 (0.0113)	0.0460 (0.0094)	0.0460 (0.0094)	100	0.0000 (0.0000)	0.0520 (0.0099)	0.0380 (0.0086)	0.0260 (0.0071)	100	0.0000 (0.0000)	0.0460 (0.0094)	0.0500 (0.0097)	0.0500 (0.0097)
200	0.0000 (0.0000)	0.0520 (0.0099)	0.0440 (0.0092)	0.0280 (0.0071)	200	0.0000 (0.0000)	0.0320 (0.0079)	0.0760 (0.0119)	0.0480 (0.0096)	200	0.0000 (0.0000)	0.0360 (0.0083)	0.0400 (0.0088)	0.0540 (0.0101)
500	0.0000 (0.0000)	0.0400 (0.0088)	0.0380 (0.0086)	0.0360 (0.0083)	500	0.0000 (0.0000)	0.0540 (0.0101)	0.0560 (0.0103)	0.0240 (0.0068)	500	0.0000 (0.0000)	0.0640 (0.0109)	0.0520 (0.0099)	0.0500 (0.0097)
1000	0.0000 (0.0000)	0.0540 (0.0101)	0.0600 (0.0106)	0.0160 (0.0056)	1000	0.0000 (0.0000)	0.0480 (0.0096)	0.0680 (0.0113)	0.0340 (0.0081)	1000	0.0000 (0.0000)	0.0820 (0.0123)	0.0340 (0.0081)	0.0580 (0.0105)

Table 58: Simulation VIII: Empirical power and Monte Carlo standard error for Localized alternative: $x_0 = 0.75$, $\delta = 0.2$, $c = 0.5$, restricted to $r = 8, 10, 15$.

$r = 8$	Uniform	Central	Tail	Localized	$r = 10$	Uniform	Central	Tail	Localized	$r = 15$	Uniform	Central	Tail	Localized
50	0.0000 (0.0000)	0.0540 (0.0101)	0.0900 (0.0128)	0.0560 (0.0103)	50	0.0000 (0.0000)	0.0820 (0.0123)	0.0540 (0.0101)	0.0420 (0.0090)	50	0.0000 (0.0000)	0.0460 (0.0094)	0.0540 (0.0101)	0.0740 (0.0117)
100	0.0000 (0.0000)	0.0460 (0.0094)	0.0360 (0.0083)	0.0420 (0.0090)	100	0.0000 (0.0000)	0.0720 (0.0116)	0.0500 (0.0097)	0.0460 (0.0094)	100	0.0000 (0.0000)	0.0600 (0.0106)	0.0500 (0.0097)	0.0500 (0.0097)
200	0.0000 (0.0000)	0.0580 (0.0105)	0.0500 (0.0097)	0.0300 (0.0076)	200	0.0000 (0.0000)	0.0560 (0.0103)	0.0660 (0.0111)	0.0580 (0.0105)	200	0.0000 (0.0000)	0.0400 (0.0088)	0.0520 (0.0099)	0.0380 (0.0086)
500	0.0000 (0.0000)	0.0600 (0.0106)	0.0580 (0.0105)	0.0180 (0.0059)	500	0.0000 (0.0000)	0.0280 (0.0074)	0.0780 (0.0120)	0.0520 (0.0099)	500	0.0000 (0.0000)	0.0300 (0.0076)	0.0380 (0.0086)	0.0340 (0.0081)
1000	0.0000 (0.0000)	0.0380 (0.0086)	0.0620 (0.0108)	0.0440 (0.0092)	1000	0.0000 (0.0000)	0.0460 (0.0094)	0.0520 (0.0099)	0.0480 (0.0096)	1000	0.0000 (0.0000)	0.0580 (0.0105)	0.0440 (0.0092)	0.0600 (0.0106)

Table 59: Simulation VIII: Empirical power and Monte Carlo standard error for Localized alternative: $x_0 = 0.75$, $\delta = 0.2$, $c = 1.5$, restricted to $r = 8, 10, 15$.

$r = 8$	Uniform	Central	Tail	Localized	$r = 10$	Uniform	Central	Tail	Localized	$r = 15$	Uniform	Central	Tail	Localized
50	0.0000 (0.0000)	0.0540 (0.0101)	0.0540 (0.0101)	0.0440 (0.0092)	50	0.0000 (0.0000)	0.0760 (0.0119)	0.0480 (0.0096)	0.0460 (0.0094)	50	0.0000 (0.0000)	0.0460 (0.0094)	0.0480 (0.0096)	0.0540 (0.0101)
100	0.0000 (0.0000)	0.0560 (0.0103)	0.0400 (0.0088)	0.0280 (0.0074)	100	0.0000 (0.0000)	0.0380 (0.0086)	0.0440 (0.0092)	0.0280 (0.0074)	100	0.0000 (0.0000)	0.0280 (0.0074)	0.0620 (0.0108)	0.0300 (0.0076)
200	0.0000 (0.0000)	0.0340 (0.0081)	0.0500 (0.0097)	0.0200 (0.0063)	200	0.0000 (0.0000)	0.0580 (0.0105)	0.0580 (0.0105)	0.0340 (0.0081)	200	0.0000 (0.0000)	0.0320 (0.0079)	0.0440 (0.0092)	0.0380 (0.0086)
500	0.0000 (0.0000)	0.0620 (0.0108)	0.0500 (0.0097)	0.0360 (0.0083)	500	0.0000 (0.0000)	0.0520 (0.0099)	0.0680 (0.0113)	0.0500 (0.0097)	500	0.0000 (0.0000)	0.0460 (0.0094)	0.0440 (0.0092)	0.0220 (0.0066)
1000	0.0000 (0.0000)	0.0540 (0.0101)	0.0640 (0.0109)	0.0400 (0.0088)	1000	0.0000 (0.0000)	0.0300 (0.0076)	0.0560 (0.0103)	0.0340 (0.0081)	1000	0.0000 (0.0000)	0.0720 (0.0116)	0.0420 (0.0090)	0.0280 (0.0074)

Table 60: Simulation VIII: Empirical power and Monte Carlo standard error for Abrupt alternative: $x_0 = 0.25$, $\delta = 0.05$, restricted to $r = 8, 10, 15$.

$r = 8$	Uniform	Central	Tail	$r = 10$	Uniform	Central	Tail	$r = 15$	Uniform	Central	Tail
50	0.0000 (0.0000)	0.0500 (0.0097)	0.0780 (0.0120)	50	0.0000 (0.0000)	0.0860 (0.0125)	0.0440 (0.0092)	50	0.0000 (0.0000)	0.0540 (0.0101)	0.0560 (0.0103)
100	0.0000 (0.0000)	0.0780 (0.0120)	0.0500 (0.0097)	100	0.0000 (0.0000)	0.0500 (0.0097)	0.0560 (0.0103)	100	0.0000 (0.0000)	0.0340 (0.0081)	0.0680 (0.0113)
200	0.0000 (0.0000)	0.0880 (0.0127)	0.0460 (0.0094)	200	0.0000 (0.0000)	0.0440 (0.0092)	0.0420 (0.0090)	200	0.0000 (0.0000)	0.0400 (0.0088)	0.0400 (0.0088)
500	0.0000 (0.0000)	0.0640 (0.0109)	0.0440 (0.0092)	500	0.0000 (0.0000)	0.0500 (0.0097)	0.0520 (0.0099)	500	0.0000 (0.0000)	0.0460 (0.0094)	0.0560 (0.0103)
1000	0.0000 (0.0000)	0.0660 (0.0111)	0.0460 (0.0094)	1000	0.0000 (0.0000)	0.0560 (0.0103)	0.0440 (0.0092)	1000	0.0000 (0.0000)	0.0760 (0.0119)	0.0400 (0.0088)

Table 61: Simulation VIII: Empirical power and Monte Carlo standard error for Abrupt alternative: $x_0 = 0.25$, $\delta = 0.2$, restricted to $r = 8, 10, 15$.

$r = 8$	Uniform	Central	Tail	$r = 10$	Uniform	Central	Tail	$r = 15$	Uniform	Central	Tail
50	0.0000 (0.0000)	0.0320 (0.0079)	0.0900 (0.0128)	50	0.0000 (0.0000)	0.0620 (0.0108)	0.0700 (0.0114)	50	0.0000 (0.0000)	0.0260 (0.0071)	0.0800 (0.0121)
100	0.0000 (0.0000)	0.0580 (0.0105)	0.0500 (0.0097)	100	0.0000 (0.0000)	0.0580 (0.0105)	0.0600 (0.0106)	100	0.0000 (0.0000)	0.0140 (0.0053)	0.0840 (0.0124)
200	0.0000 (0.0000)	0.0400 (0.0088)	0.0640 (0.0109)	200	0.0000 (0.0000)	0.0420 (0.0090)	0.0820 (0.0123)	200	0.0000 (0.0000)	0.0100 (0.0044)	0.0660 (0.0111)
500	0.0000 (0.0000)	0.0400 (0.0088)	0.0500 (0.0097)	500	0.0000 (0.0000)	0.0360 (0.0083)	0.0800 (0.0121)	500	0.0000 (0.0000)	0.0200 (0.0063)	0.0780 (0.0120)
1000	0.0000 (0.0000)	0.0440 (0.0092)	0.0500 (0.0097)	1000	0.0000 (0.0000)	0.0380 (0.0086)	0.0760 (0.0119)	1000	0.0000 (0.0000)	0.0300 (0.0076)	0.0740 (0.0117)

Table 62: Simulation VIII: Empirical power and Monte Carlo standard error for Abrupt alternative: $x_0 = 0.5$, $\delta = 0.1$, restricted to $r = 8, 10, 15$.

$r = 8$	Uniform	Central	Tail	$r = 10$	Uniform	Central	Tail	$r = 15$	Uniform	Central	Tail
50	0.0000 (0.0000)	0.0520 (0.0099)	0.0820 (0.0123)	50	0.0000 (0.0000)	0.0780 (0.0120)	0.0560 (0.0103)	50	0.0000 (0.0000)	0.0500 (0.0097)	0.0480 (0.0094)
100	0.0000 (0.0000)	0.0520 (0.0099)	0.0700 (0.0114)	100	0.0000 (0.0000)	0.0680 (0.0113)	0.0420 (0.0090)	100	0.0000 (0.0000)	0.0540 (0.0101)	0.0600 (0.0106)
200	0.0000 (0.0000)	0.0520 (0.0099)	0.0560 (0.0103)	200	0.0000 (0.0000)	0.0680 (0.0113)	0.0660 (0.0111)	200	0.0000 (0.0000)	0.0420 (0.0090)	0.0500 (0.0097)
500	0.0000 (0.0000)	0.0540 (0.0101)	0.0620 (0.0108)	500	0.0000 (0.0000)	0.0500 (0.0097)	0.0800 (0.0121)	500	0.0000 (0.0000)	0.0560 (0.0103)	0.0580 (0.0105)
1000	0.0000 (0.0000)	0.0400 (0.0088)	0.0580 (0.0105)	1000	0.0000 (0.0000)	0.0560 (0.0103)	0.0580 (0.0105)	1000	0.0000 (0.0000)	0.0840 (0.0124)	0.0640 (0.0109)

Table 63: Simulation VIII: Empirical power and Monte Carlo standard error for Abrupt alternative: $x_0 = 0.75$, $\delta = 0.05$, restricted to $r = 8, 10, 15$.

$r = 8$	Uniform	Central	Tail	$r = 10$	Uniform	Central	Tail	$r = 15$	Uniform	Central	Tail
50	0.0000 (0.0000)	0.0460 (0.0094)	0.0580 (0.0105)	50	0.0000 (0.0000)	0.0840 (0.0124)	0.0460 (0.0094)	50	0.0000 (0.0000)	0.0420 (0.0090)	0.0460 (0.0094)
100	0.0000 (0.0000)	0.0440 (0.0092)	0.0560 (0.0103)	100	0.0000 (0.0000)	0.0560 (0.0103)	0.0460 (0.0094)	100	0.0000 (0.0000)	0.0380 (0.0086)	0.0420 (0.0090)
200	0.0000 (0.0000)	0.0200 (0.0063)	0.0740 (0.0117)	200	0.0000 (0.0000)	0.0180 (0.0059)	0.1020 (0.0135)	200	0.0000 (0.0000)	0.0280 (0.0074)	0.0720 (0.0116)
500	0.0000 (0.0000)	0.0080 (0.0040)	0.1140 (0.0142)	500	0.0000 (0.0000)	0.0200 (0.0063)	0.1000 (0.0134)	500	0.0000 (0.0000)	0.0120 (0.0049)	0.1600 (0.0164)
1000	0.0000 (0.0000)	0.0080 (0.0040)	0.1740 (0.0170)	1000	0.0000 (0.0000)	0.0120 (0.0049)	0.1400 (0.0155)	1000	0.0000 (0.0000)	0.0180 (0.0059)	0.1560 (0.0162)

Table 64: Simulation VIII: Empirical power and Monte Carlo standard error for Abrupt alternative: $x_0 = 0.75$, $\delta = 0.2$, restricted to $r = 8, 10, 15$.

$r = 8$	Uniform	Central	Tail	$r = 10$	Uniform	Central	Tail	$r = 15$	Uniform	Central	Tail
50	0.0000 (0.0000)	0.0180 (0.0059)	0.0700 (0.0114)	50	0.0000 (0.0000)	0.0440 (0.0092)	0.0320 (0.0079)	50	0.0000 (0.0000)	0.0500 (0.0097)	0.0340 (0.0081)
100	0.0000 (0.0000)	0.0020 (0.0020)	0.2180 (0.0185)	100	0.0000 (0.0000)	0.0000 (0.0000)	0.1260 (0.0148)	100	0.0000 (0.0000)	0.0020 (0.0020)	0.0640 (0.0109)
200	0.0000 (0.0000)	0.0000 (0.0000)	0.6040 (0.0219)	200	0.0000 (0.0000)	0.0000 (0.0000)	0.5900 (0.0220)	200	0.0000 (0.0000)	0.0000 (0.0000)	0.3400 (0.0212)
500	0.0000 (0.0000)	0.0000 (0.0000)	0.9460 (0.0101)	500	0.0000 (0.0000)	0.0000 (0.0000)	0.9800 (0.0063)	500	0.0000 (0.0000)	0.0000 (0.0000)	0.9720 (0.0074)
1000	0.0000 (0.0000)	0.0000 (0.0000)	1.0000 (0.0000)	1000	0.0000 (0.0000)	0.0000 (0.0000)	1.0000 (0.0000)	1000	0.0000 (0.0000)	0.0000 (0.0000)	1.0000 (0.0000)

Table 65: Simulation VIII: Empirical power and Monte Carlo standard error for Oscillatory alternative: $x_0 = 1$, $\delta = 0.5$, restricted to $r = 8, 10, 15$.

$r = 8$	Uniform	Central	Tail	$r = 10$	Uniform	Central	Tail	$r = 15$	Uniform	Central	Tail
50	0.0000 (0.0000)	0.0460 (0.0094)	0.0640 (0.0109)	50	0.0000 (0.0000)	0.0680 (0.0113)	0.0300 (0.0076)	50	0.0000 (0.0000)	0.0520 (0.0099)	0.0460 (0.0094)
100	0.0000 (0.0000)	0.0520 (0.0099)	0.0480 (0.0096)	100	0.0000 (0.0000)	0.0560 (0.0103)	0.0520 (0.0099)	100	0.0000 (0.0000)	0.0280 (0.0074)	0.0460 (0.0094)
200	0.0000 (0.0000)	0.0520 (0.0099)	0.0640 (0.0109)	200	0.0000 (0.0000)	0.0380 (0.0086)	0.0600 (0.0106)	200	0.0000 (0.0000)	0.0320 (0.0079)	0.0540 (0.0101)
500	0.0000 (0.0000)	0.0540 (0.0101)	0.0460 (0.0094)	500	0.0000 (0.0000)	0.0260 (0.0071)	0.0660 (0.0111)	500	0.0000 (0.0000)	0.0400 (0.0088)	0.0660 (0.0111)
1000	0.0000 (0.0000)	0.0400 (0.0088)	0.0500 (0.0097)	1000	0.0000 (0.0000)	0.0560 (0.0103)	0.0600 (0.0106)	1000	0.0000 (0.0000)	0.0640 (0.0109)	0.0400 (0.0088)

Table 66: Simulation VIII: Empirical power and Monte Carlo standard error for Oscillatory alternative: $x_0 = 1$, $\delta = 1.5$, restricted to $r = 8, 10, 15$.

$r = 8$	Uniform	Central	Tail	$r = 10$	Uniform	Central	Tail	$r = 15$	Uniform	Central	Tail
50	0.0000 (0.0000)	0.0500 (0.0097)	0.0640 (0.0109)	50	0.0000 (0.0000)	0.0740 (0.0117)	0.0320 (0.0079)	50	0.0000 (0.0000)	0.0480 (0.0096)	0.0540 (0.0101)
100	0.0000 (0.0000)	0.0600 (0.0106)	0.0420 (0.0090)	100	0.0000 (0.0000)	0.0580 (0.0105)	0.0440 (0.0092)	100	0.0000 (0.0000)	0.0400 (0.0088)	0.0740 (0.0117)
200	0.0000 (0.0000)	0.0380 (0.0086)	0.0520 (0.0099)	200	0.0000 (0.0000)	0.0380 (0.0086)	0.0640 (0.0109)	200	0.0000 (0.0000)	0.0420 (0.0090)	0.0480 (0.0096)
500	0.0000 (0.0000)	0.0500 (0.0097)	0.0300 (0.0076)	500	0.0000 (0.0000)	0.0460 (0.0094)	0.0620 (0.0108)	500	0.0000 (0.0000)	0.0440 (0.0092)	0.0740 (0.0117)
1000	0.0000 (0.0000)	0.0400 (0.0088)	0.0460 (0.0094)	1000	0.0000 (0.0000)	0.0320 (0.0079)	0.0580 (0.0105)	1000	0.0000 (0.0000)	0.0660 (0.0111)	0.0560 (0.0103)

Table 67: Simulation VIII: Empirical power and Monte Carlo standard error for Oscillatory alternative: $x_0 = 2$, $\delta = 0.5$, restricted to $r = 8, 10, 15$.

$r = 8$	Uniform	Central	Tail	$r = 10$	Uniform	Central	Tail	$r = 15$	Uniform	Central	Tail
50	0.0000 (0.0000)	0.0580 (0.0105)	0.0840 (0.0124)	50	0.0000 (0.0000)	0.0900 (0.0128)	0.0500 (0.0097)	50	0.0000 (0.0000)	0.0420 (0.0090)	0.0440 (0.0092)
100	0.0000 (0.0000)	0.0520 (0.0099)	0.0500 (0.0097)	100	0.0000 (0.0000)	0.0740 (0.0117)	0.0480 (0.0096)	100	0.0000 (0.0000)	0.0400 (0.0088)	0.0600 (0.0106)
200	0.0000 (0.0000)	0.0580 (0.0105)	0.0560 (0.0103)	200	0.0000 (0.0000)	0.0540 (0.0101)	0.0540 (0.0101)	200	0.0000 (0.0000)	0.0440 (0.0092)	0.0520 (0.0099)
500	0.0000 (0.0000)	0.0800 (0.0121)	0.0440 (0.0092)	500	0.0000 (0.0000)	0.0500 (0.0097)	0.0460 (0.0094)	500	0.0000 (0.0000)	0.0460 (0.0094)	0.0700 (0.0114)
1000	0.0000 (0.0000)	0.0560 (0.0103)	0.0500 (0.0097)	1000	0.0000 (0.0000)	0.0340 (0.0081)	0.0520 (0.0099)	1000	0.0000 (0.0000)	0.0640 (0.0109)	0.0640 (0.0109)

Table 68: Simulation VIII: Empirical power and Monte Carlo standard error for Oscillatory alternative: $x_0 = 2$, $\delta = 1.5$, restricted to $r = 8, 10, 15$.

$r = 8$	Uniform	Central	Tail	$r = 10$	Uniform	Central	Tail	$r = 15$	Uniform	Central	Tail
50	0.0000 (0.0000)	0.0460 (0.0094)	0.0640 (0.0109)	50	0.0000 (0.0000)	0.0740 (0.0117)	0.0420 (0.0090)	50	0.0000 (0.0000)	0.0520 (0.0099)	0.0580 (0.0105)
100	0.0000 (0.0000)	0.0420 (0.0090)	0.0420 (0.0090)	100	0.0000 (0.0000)	0.0720 (0.0116)	0.0440 (0.0092)	100	0.0000 (0.0000)	0.0340 (0.0081)	0.0520 (0.0099)
200	0.0000 (0.0000)	0.0600 (0.0106)	0.0680 (0.0113)	200	0.0000 (0.0000)	0.0480 (0.0096)	0.0700 (0.0114)	200	0.0000 (0.0000)	0.0380 (0.0086)	0.0780 (0.0120)
500	0.0000 (0.0000)	0.0400 (0.0088)	0.0340 (0.0081)	500	0.0000 (0.0000)	0.0380 (0.0086)	0.0640 (0.0109)	500	0.0000 (0.0000)	0.0440 (0.0092)	0.0720 (0.0116)
1000	0.0000 (0.0000)	0.0300 (0.0076)	0.0540 (0.0101)	1000	0.0000 (0.0000)	0.0360 (0.0083)	0.0720 (0.0116)	1000	0.0000 (0.0000)	0.0820 (0.0123)	0.0520 (0.0099)

Table 69: Simulation VIII: Empirical power and Monte Carlo standard error for Oscillatory alternative: $x_0 = 4$, $\delta = 0.5$, restricted to $r = 8, 10, 15$.

$r = 8$	Uniform	Central	Tail	$r = 10$	Uniform	Central	Tail	$r = 15$	Uniform	Central	Tail
50	0.0000 (0.0000)	0.0480 (0.0096)	0.0620 (0.0108)	50	0.0000 (0.0000)	0.0940 (0.0131)	0.0460 (0.0094)	50	0.0000 (0.0000)	0.0460 (0.0094)	0.0600 (0.0106)
100	0.0000 (0.0000)	0.0400 (0.0088)	0.0440 (0.0092)	100	0.0000 (0.0000)	0.0740 (0.0117)	0.0500 (0.0097)	100	0.0000 (0.0000)	0.0440 (0.0092)	0.0380 (0.0086)
200	0.0000 (0.0000)	0.0460 (0.0094)	0.0540 (0.0101)	200	0.0000 (0.0000)	0.0420 (0.0090)	0.0500 (0.0097)	200	0.0000 (0.0000)	0.0580 (0.0105)	0.0520 (0.0099)
500	0.0000 (0.0000)	0.0360 (0.0083)	0.0460 (0.0094)	500	0.0000 (0.0000)	0.0300 (0.0076)	0.0560 (0.0103)	500	0.0000 (0.0000)	0.0540 (0.0101)	0.0320 (0.0079)
1000	0.0000 (0.0000)	0.0300 (0.0076)	0.0720 (0.0116)	1000	0.0000 (0.0000)	0.0520 (0.0099)	0.0460 (0.0094)	1000	0.0000 (0.0000)	0.0600 (0.0106)	0.0560 (0.0103)

Table 70: Simulation VIII: Empirical power and Monte Carlo standard error for Oscillatory alternative: $x_0 = 4$, $\delta = 1.5$, restricted to $r = 8, 10, 15$.

$r = 8$	Uniform	Central	Tail	$r = 10$	Uniform	Central	Tail	$r = 15$	Uniform	Central	Tail
50	0.0000 (0.0000)	0.0380 (0.0086)	0.0780 (0.0120)	50	0.0000 (0.0000)	0.0780 (0.0120)	0.0400 (0.0088)	50	0.0000 (0.0000)	0.0540 (0.0101)	0.0380 (0.0086)
100	0.0000 (0.0000)	0.0520 (0.0099)	0.0540 (0.0101)	100	0.0000 (0.0000)	0.0580 (0.0105)	0.0580 (0.0105)	100	0.0000 (0.0000)	0.0440 (0.0092)	0.0380 (0.0086)
200	0.0000 (0.0000)	0.0600 (0.0106)	0.0640 (0.0109)	200	0.0000 (0.0000)	0.0500 (0.0097)	0.0800 (0.0121)	200	0.0000 (0.0000)	0.0440 (0.0092)	0.0420 (0.0090)
500	0.0000 (0.0000)	0.0500 (0.0097)	0.0260 (0.0071)	500	0.0000 (0.0000)	0.0340 (0.0081)	0.0660 (0.0111)	500	0.0000 (0.0000)	0.0380 (0.0086)	0.0820 (0.0123)
1000	0.0000 (0.0000)	0.0520 (0.0099)	0.0600 (0.0106)	1000	0.0000 (0.0000)	0.0360 (0.0083)	0.0520 (0.0099)	1000	0.0000 (0.0000)	0.0520 (0.0099)	0.0400 (0.0088)

Table 71: Simulation VIII: Empirical power and Monte Carlo standard error for Oscillatory alternative: $x_0 = 8$, $\delta = 0.5$, restricted to $r = 8, 10, 15$.

$r = 8$	Uniform	Central	Tail	$r = 10$	Uniform	Central	Tail	$r = 15$	Uniform	Central	Tail
50	0.0000 (0.0000)	0.0520 (0.0099)	0.0500 (0.0097)	50	0.0000 (0.0000)	0.0840 (0.0124)	0.0480 (0.0096)	50	0.0000 (0.0000)	0.0500 (0.0097)	0.0340 (0.0081)
100	0.0000 (0.0000)	0.0520 (0.0099)	0.0480 (0.0096)	100	0.0000 (0.0000)	0.0620 (0.0108)	0.0380 (0.0086)	100	0.0000 (0.0000)	0.0420 (0.0090)	0.0520 (0.0099)
200	0.0000 (0.0000)	0.0460 (0.0094)	0.0580 (0.0105)	200	0.0000 (0.0000)	0.0440 (0.0092)	0.0600 (0.0106)	200	0.0000 (0.0000)	0.0300 (0.0076)	0.0480 (0.0096)
500	0.0000 (0.0000)	0.0600 (0.0106)	0.0380 (0.0086)	500	0.0000 (0.0000)	0.0460 (0.0094)	0.0740 (0.0117)	500	0.0000 (0.0000)	0.0320 (0.0079)	0.0700 (0.0114)
1000	0.0000 (0.0000)	0.0480 (0.0096)	0.0560 (0.0103)	1000	0.0000 (0.0000)	0.0380 (0.0086)	0.0680 (0.0113)	1000	0.0000 (0.0000)	0.0520 (0.0099)	0.0400 (0.0088)

Table 73: Simulation VIII: Empirical power and Monte Carlo standard error for Mixture alternative: $x_0 = 0.05$, $\delta = 0.5$, restricted to $r = 8, 10, 15$.

Table 74: Simulation VIII: Empirical power and Monte Carlo standard error for Mixture alternative: $x_0 = 0.1$, $\delta = 0.25$, restricted to $r = 8, 10, 15$.

Table 75: Simulation VIII: Empirical power and Monte Carlo standard error for Mixture alternative: $x_0 = 0.1$, $\delta = 0.75$, restricted to $r = 8, 10, 15$.

Table 76: Simulation VIII: Empirical power and Monte Carlo standard error for Mixture alternative: $x_0 = 0.2$, $\delta = 0.5$, restricted to $r = 8, 10, 15$.